State density III: positivity and the asymptotic-equivalence form $Z(n) \sim C(\xi, \eta)\, n^{-p/2}$

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Abstract

The divisor density of unit 70 is measurable and integrable on the box, and positive when $\eta(0, \cdot) > 0$, so its integral $C(\xi, \eta) = \frac{1}{k_0} \int v^{h'} (v^{k'})^{-p} A_{p-1}(\xi(0, v))\eta(0, v)\, dv$ is a positive constant and the state integral satisfies $Z(n) \sim C(\xi, \eta)\, n^{-p/2}$. Along the way: the fluctuation mass $A_\gamma(a)$ is monotone, hence measurable, in a. Zero `sorry`/`axiom`.

1 The things formalised

```
theorem weightedMass_mono_left (    :  ) (h  : 0 <  ) (h  : -1 <  ) :
    Monotone fun a => weightedMass   a
theorem divisorWeight_integrable ... (hq :   i, p < ((h' i :  ) + 1) / k' i) :
    Integrable (fun v =>   i, (v i ^ h' i * (v i ^ k' i) ^ (-p))) (boxMeasure b d)
theorem divisorDensity_integrable ... :
    Integrable (divisorDensity   h  k  k' h'    ) (boxMeasure b d)
theorem integral_divisorDensity_pos ...
    (h pos :   v, (  i, 0 < v i   v i   b) → 0 <   0 v) :
    0 <   v, divisorDensity   h  k  k' h'    v  (boxMeasure b d)
theorem stateIntegral_isEquivalent ... :
    (fun n => stateIntegral   b n h  k  k' h'    ) ~[atTop]
      fun n => (  v, divisorDensity   h  k  k' h'    v  (boxMeasure b d))
        * n ^ (-(((h  :  ) + 1) / k  / 2))
```

2 Proof strategy

Monotonicity of $a \mapsto A_\gamma(a)$ is pointwise monotonicity of $e^{\beta a t}$; measurability follows. The density is dominated by $K \prod_i v_i^{h'_i} (v_i^{k'_i})^{-p}$ with K built from the compact bounds on $\xi(0, \cdot), \eta(0, \cdot)$ and the monotonicity of A; the weight is a product of single-coordinate integrable functions. Positivity is `integral_pos_iff_support_of_nonneg_ae` with the whole box inside the support and `boxMeasure`(box) $=$ (ofReal $b)^d > 0$ (`Measure.pi_pi`). The equivalence multiplies the limit of $n^{p/2} Z$ by $n^{-p/2}$.

3 Roadblocks and resolutions

- A bare `by` block as the left argument of `.trans` has no expected type; state the intermediate fact with `have`.
- The `rpow` exponent identity closes with `ring_nf` rather than `congr 1; ring`.

4 Outcome

For a unique minimal coordinate the state-density coefficient is positive and gives a sharp asymptotic equivalence. Remaining in Programme A: the Fubini bridge to the $(d+1)$-dimensional box; then Programme B (several minimal coordinates).